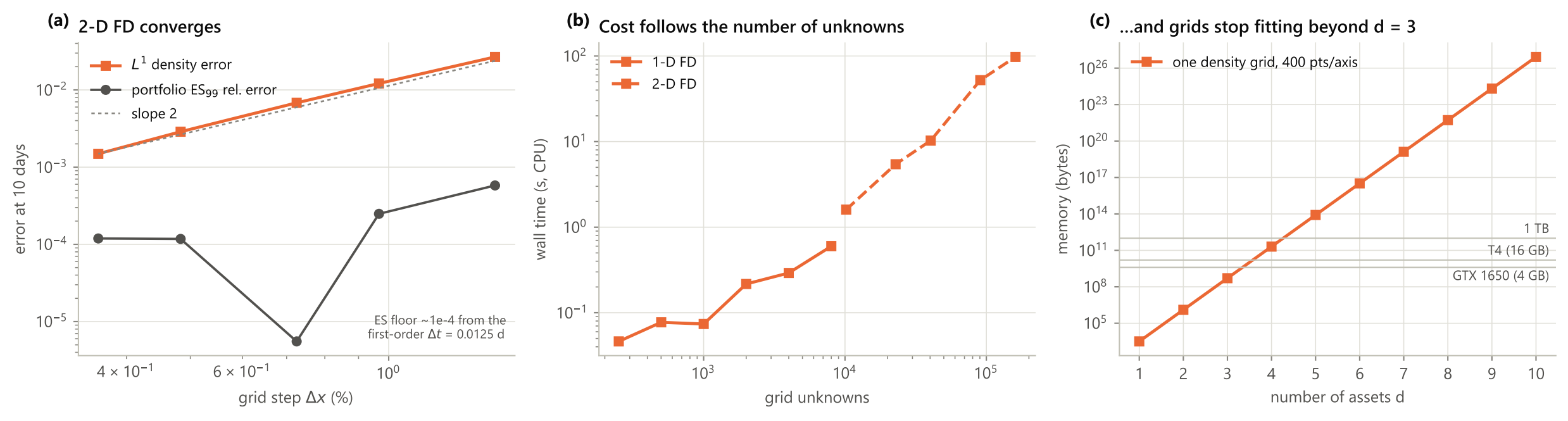
(a) 2-D FD converges
L^1 density error
portfolio ES_{99} rel. error
slope 2
ES floor ~1e-4 from the first-order Δt = 0.0125 d
error at 10 days
grid step Δx (%)
(b) Cost follows the number of unknowns
1-D FD
2-D FD
wall time (s, CPU)
grid unknowns
(c) ...and grids stop fitting beyond d = 3
one density grid, 400 pts/axis
1 TB
T4 (16 GB)
GTX 1650 (4 GB)
memory (bytes)
number of assets d